GMIU – Gyanmanjari Innovative University

Activity Report

Malware Behaviour Simulator

Behaviour and Impact of Viruses, Worms and Trojans

Detail	Information
Student Name	Priyrajsinh
Course	DETIT15214 – Cryptography and Network Security
Session	GMIU-2026-27 (ODD)
Semester / Section	Sem 5 / Class B
Live Simulation	https://cns-ala-2-priyrajsinh.netlify.app

1. Introduction

Malware (malicious software) is any program designed to damage, disrupt or gain unauthorised access to a computer system. The three classic categories are viruses, worms and trojans. They differ mainly in how they enter a system and how they spread. Understanding these differences is a basic requirement of network security, because the defence against each type is different.

This activity presents a safe, controlled simulation that demonstrates how these three threats behave inside a small network of 12 computers connected through a central switch. The simulation is purely visual. It does not access real files, processes, networks or operating-system resources, and it contains no real malware code.

2. Objective

- To demonstrate the behaviour of a virus, a worm and a trojan in a safe and controlled environment.
- To show visually how each threat spreads, replicates or disguises itself.
- To analyse the impact of each type of malware on a network.
- To explain suitable prevention and protection measures.

3. About the Simulator

The simulator is a web page built with HTML, CSS and JavaScript. It is hosted online and also works offline as a single file. It shows a network of 12 PCs arranged around a central switch. Dashed lines between neighbouring PCs represent file sharing (USB drives, e-mail attachments), while solid lines represent LAN cables connected to the switch.

3.1 Features

- Selection of the threat: Virus, Worm or Trojan.
- Animated network where infected PCs turn red, glow and shake, and ripples spread from them.
- Moving packets that show files or malware travelling between computers.
- Live counters for infected PCs, number of events and current status, plus an infection progress bar.
- A live activity log that explains every step in plain language.
- Information panels: properties, working flow, solution, detailed information and security recommendation.
- Alert banner and red screen flash when the whole network is compromised or a backdoor opens.
- Change Threat and Restart buttons to repeat the experiment.

3.2 Safety

The program only changes colours, text and shapes on the web page. It never reads, writes or deletes any file, never sends any network request, and never runs any system command. Therefore it is completely harmless and suitable for classroom demonstration.

4. Virus

4.1 Definition

A computer virus is a malicious program that attaches itself to a legitimate host file or program. It stays inactive until the infected file is opened or executed by a user. After activation it inserts copies of itself into other files.

4.2 How it spreads

- Infected USB drives and other removable media.
- E-mail attachments such as documents with malicious macros.
- Shared folders and downloaded files.

4.3 Behaviour in the simulation

One random PC opens an infected file and turns red. After some time that PC shares a file with a neighbouring PC along a dashed link. The file travels as a moving packet, and only when the neighbour opens it does that PC become infected. Because every step needs a human action, the infection spreads slowly and step by step.

4.4 Impact

- Corruption or deletion of files and loss of important data.
- Slower system performance and unexpected crashes.
- Infected files can silently reach other users, colleagues and customers.
- Time and money spent on cleaning and recovery.

4.5 Prevention

- Use updated antivirus software and scan USB drives and attachments before opening.
- Disable auto-run for removable drives and block macros from unknown documents.
- Keep regular, offline backups of important data.

5. Worm

5.1 Definition

A worm is a standalone malicious program that copies itself from one computer to another through a network, usually by exploiting a security vulnerability. It does not need a host file and does not need any action from the user.

5.2 How it spreads

- Scanning the network for computers with unpatched software or open ports.
- Sending copies of itself through network connections, shared folders or e-mail.
- Every newly infected computer starts scanning and infecting others, so growth is exponential.

5.3 Behaviour in the simulation

One PC is infected automatically through a vulnerability. Green radar ripples show it scanning the LAN. Copies of the worm travel through the central switch and infect several PCs at the same time. Each new victim joins the scanning, so within a few seconds all 12 PCs are infected and a "Network Compromised" alert appears. The difference in speed compared with the virus is the key lesson of this demonstration.

5.4 Impact

- Heavy consumption of network bandwidth and system resources, which can bring down the whole network.
- Very fast spread, so there is little time to react.
- Worms often carry a payload such as ransomware or a backdoor.
- Business interruption and large financial loss.

5.5 Prevention

- Install operating system and software security patches quickly.
- Use firewalls and close unused ports and services.
- Segment the network so that an infection cannot reach every computer.
- Use intrusion detection and prevention systems to notice abnormal scanning traffic.

6. Trojan

6.1 Definition

A trojan horse is malware that pretends to be useful or harmless software, such as a free game, a cracked application or a fake update. The user installs it willingly, but it carries a hidden malicious function. A trojan does not replicate itself.

6.2 How it spreads

- Fake or cracked software on untrusted download sites.
- Malicious e-mail attachments and links that look genuine.
- Bundled with free programs that the user installs without checking.

6.3 Behaviour in the simulation

An attacker uploads a fake file named free_game.exe. A single PC downloads it and a normal-looking installer window with a progress bar appears, so the PC looks safe (yellow). After the installer closes, the hidden payload becomes active and the PC turns red. A red dashed backdoor line then connects the PC to the attacker, and data packets flow towards the attacker to show data theft. The other 11 PCs stay clean, which demonstrates that a trojan targets specific victims and does not spread by itself.

6.4 Impact

- Backdoor access that lets the attacker control the computer remotely.
- Theft of passwords, banking details, personal files and keystrokes.
- The infected computer can be used to download more malware or to attack others.
- Loss of privacy and identity or financial fraud.

6.5 Prevention

- Download software only from official and trusted sources and avoid cracked programs.
- Verify digital signatures or file hashes where available.
- Use least-privilege user accounts, application whitelisting and endpoint protection.
- Use a firewall that blocks unknown outbound connections.

7. Comparison of Virus, Worm and Trojan

Feature	Virus	Worm	Trojan
Needs host file	Yes	No (standalone)	No (disguised program)
User action needed	Yes, opens infected file	No	Yes, user installs it
Self-replication	Yes, into other files	Yes, across the network	No
Speed of spread	Slow	Very fast	No spread (single target)
Main medium	USB, e-mail, shared files	Network / LAN	Fake downloads, e-mail
Main damage	File corruption, data loss	Network overload, payloads	Backdoor, data theft
Main defence	Antivirus, safe file handling	Patching, firewall, segmentation	Trusted sources, whitelisting

8. Observations from the Simulation

- The virus infected PCs one at a time, because each infection depended on a user action.
- The worm infected the whole network in a few seconds, because it spread automatically and in parallel.
- The trojan infected only one PC, but that PC gave the attacker complete remote access.
- The simplest protections, which are updates, trusted software and user awareness, defend against all three threats.

9. General Security Measures

- Keep the operating system, applications and antivirus up to date.
- Use a firewall and, in organisations, network segmentation and intrusion detection.
- Take regular backups and keep a copy offline.
- Use strong passwords and multi-factor authentication.
- Train users not to open unknown attachments or run unknown software.
- Apply the principle of least privilege to all accounts.

10. Conclusion

The simulator clearly shows that viruses, worms and trojans are different in the way they enter and spread. A virus depends on a host file and a user, a worm spreads by itself across the network, and a trojan tricks the user into installing it and then opens a backdoor. Because each one has a different weakness, network security needs layered protection: updated software, firewalls, trusted sources of software, backups and user awareness. This activity helped to understand these concepts safely, without using any real malware.

11. References

- William Stallings, Cryptography and Network Security: Principles and Practice.
- Charles P. Pfleeger, Security in Computing.
- CERT-In (Indian Computer Emergency Response Team) – Guidelines on malware and cyber hygiene, cert-in.org.in.
- NIST SP 800-83, Guide to Malware Incident Prevention and Handling for Desktops and Laptops.